Table 3.4: General Properties:

	HF	HCl	HBr	HI
Bond dissolution enthalphy	+562	+431	+366	+299
% of ionic character	43	17	113	7

In line with the decreasing bond dissociation enthalpy, the thermal stability of hydrogen halides decreases from fluoride to iodide.

For example, Hydrogen iodide decomposes at 400° C while hydrogen fluoride and hydrogen chloride are stable at this temperature.

At room temperature, hydrogen halides are gases but hydrogen fluoride can be readily liquefied. The gases are colourless but, with moist air gives white fumes due to the production of droplets of hydrohalic acid. In HF, due to the presence of strong hydrogen bond it has high melting and boiling points. This effect is absent in other hydrogen halides.

Acidic properties:

The hydrogen halides are extremely soluble in water due to the ionisation.

$$HX + H_2O \longrightarrow H_3O^+ + X^-$$

(X – F, Cl, Br, or I)

Solutions of hydrogen halides are therefore acidic and known as hydrohalic acids. Hydrochloric, hydrobromic and hydroiodic acids are almost completely ionised and are therefore strong acids but HF is a weak acid i.e. 0.1mM solution is only 10% ionised, but in 5M and 15M solution HF is stronger acid due to the equilibrium.

$$HF + H_2O \rightleftharpoons H_3O^+ + F^-$$

$$HF + F^- \rightleftharpoons HF_2^-$$

At high concentration, the equilibrium involves the removal of fluoride ions is important. Since it affects the dissociation of hydrogen fluoride and increases and hydrogen ion concentration Several stable salts $NaHF_2$, KHF_2 and NH_4HF_2 are known. The other hydrogen halides do not form hydrogen dihalides.

Hydrohalic acid shows typical acidic properties. They form salts with acids, bases and reacts with metals to give hydrogen. Moist hydrofluoric acid (not dry) rapidly react with silica and glass.

$$SiO_2 + 4HF \longrightarrow SiF_4 + 2H_2O$$

$$Na_2SiO_3 + 6HF \longrightarrow Na_2SiF_6 + 3H_2O$$

Oxidation: Hydrogen iodide is readily oxidised to iodine hence it is a reducing agent.

$$2HI \rightleftharpoons 2H^+ + I_2 + 2e^-$$

Acidic solution of iodides is readily oxidised. A positive result is shown by liberation of iodine which gives a blue-black colouration with starch.

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